



# Technical Service Bulletin

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Diesel Exhaust Fluid (DEF) Quality Fault Codes on International® Truck and IC™ Bus Chassis

## Diesel Exhaust Fluid (DEF) Quality Fault Codes on International® Truck and IC™ Bus Chassis

### Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Information Only - OEM Related Matter Not Covered By Cummins® - Contact Appropriate OEM Dealer or OEM Representative For Additional Information

### Contents

#### Product Affected

Engines:

- B6.7 CM2350 B121B
- ISB6.7 CM2350 B101
- ISL9 CM2350 L101
- ISL9 CM2350 L123B
- ISX15 CM2350 X101
- L9 CM2350 L116B
- X15 CM2350 X114B
- X15 CM2350 X116B

Original equipment manufacturer (OEM):

- Navistar
- International

#### Issue

Symptom:

- Fault Code(s) 1715 (SPN 3364 FMI 11), 4277 (SPN 3364 FMI 10), 3867 (SPN 3364 FMI 18), 4842 (SPN 3364 FMI 15), 6765 (SPN 3521 FMI 16), 6766 (SPN 3521 FMI 18), 6767 (SPN 3521 FMI 31)

Root Cause:

- When the diesel exhaust fluid (DEF) level in the DEF tank is low, the DEF quality sensor is more susceptible to calculating the DEF concentration incorrectly. The miscalculation is the result of air bubbles created by the low level of DEF moving in the DEF tank.

## Verification

If the following conditions are met:

- International® ProStar, 9900, Lonestar, Paystar, HX Series, LT Series, Durastar, Workstar, or IC™ Bus
- Fault Code(s) 1715 (SPN 3364 FMI 11), 4277 (SPN 3364 FMI 10), 3867 (SPN 3364 FMI 18), 4842 (SPN 3364 FMI 15), 6765 (SPN 3521 FMI 16), 6766 (SPN 3521 FMI 18), 6767 (SPN 3521 FMI 31)

See Resolution section below for fault code being diagnosed.

## Resolution

- Reference Navistar Fault Code Action Plan (FCAP) for individual fault codes.
- Fault Code 1715 (SPN 3364 FMI 11):
  - Navistar has released an update to aftertreatment DEF quality sensor firmware to reduce occurrences of Fault Code 1715.
  - Navistar released Authorized Field Change (AFC) 16904 and 16905 to authorize update of aftertreatment DEF quality sensor firmware.
  - Authorized Field Change (AFC) **must** be completed by an authorized International® Truck or Bus Dealer. Reference Manage Repair tab within Navistar Service Portal to verify eligibility to perform update.
- Fault Code(s) 4277 (SPN 3364 FMI 10), 3867 (SPN 3364 FMI 18), 4842 (SPN 3364 FMI 15), 6765 (SPN 3521 FMI 16), 6766 (SPN 3521 FMI 18), 6767 (SPN 3521 FMI 31)
  - Navistar has released supplemental diagnostic and service information for Diesel Exhaust Fluid (DEF) Quality Sensor via iKnow Article IK0700097
  - Diagnostic Steps section in iKnow article will guide user through a series of steps to determine cause of fault code.
  - iKnow article must be followed and completed by an authorized International® Truck or Bus Dealer.

## Document History

| Date       | Details                                                     |
|------------|-------------------------------------------------------------|
| 2016-3-21  | Module Created                                              |
| 2016-10-17 | Added ISL9 CM2350 L101. Modified Resolution section.        |
| 2017-9-29  | Added more Fault Codes and products.                        |
| 2018-5-22  | Reformatted Engines and OEMs affected section.              |
| 2019-8-1   | Corrected iKnow article number to: IK0700097 from IK0700096 |

